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AQA A Level Further Mathematics (7367)

Topic **C: Matrices**

Sub topic **C8: Geometric Interpretation of Solutions**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**C: Matrices**
Sub topic	**C8: Geometric Interpretation of Solutions**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

Three planes have equations

$$x - y + kz = 3$$

$$kx - 3y + 5z = -1$$

$$x - 2y + 3z = -4$$

where k is a real constant. The planes do not meet at a unique point.

(a) Find the possible values of k

(3)

(b) There are two possible geometric configurations for the given planes.

Identify each possible configuration, stating the corresponding value of k

Fully justify your answer.

(5)

(c) Given further that the equations of the planes form a consistent system, find the solution of the system of equations.

(3)

(Total 11 marks)

Q2.

The planes Π_1, Π_2 and Π_3 have cartesian equations

$$2x + y - z = 3$$

$$3x - 2y + z = 5$$

$$12x - y - z = 40$$

respectively.

(a) Find, in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, a vector equation for the line L which is the intersection of Π_1 and Π_2.

(5)

(b) (i) Determine whether L meets Π_3 , and use your answer to decide whether the system given by the equations of these three planes is consistent or inconsistent.

(3)

(ii) Describe geometrically the arrangement of the three planes.

(1)

(c) (i) Find the coordinates of a common point of Π_2 and Π_3.

(3)

(ii) **Deduce** a vector equation for the line of intersection of Π_2 and Π_3.

(1)
(Total 13 marks)

Q3.

(a) Find the values of t for which the system of equations

$$tx + 2y + 3z = a$$

$$2x + 3y - tz = b$$

$$3x + 5y + (t + 1)z = c$$

does not have a unique solution.

(3)

(b) For the integer value of t found in part (a), find the relationship between a, b and c such that this system of equations is consistent.

(3)
(Total 6 marks)

Q4.

The system of equations S is given in terms of the real parameters a and b by

$$2x + y + 3z = a + 1$$

$$5x - 2y + (a + 1)z = 3$$

$$ax + 2y + 4z = b$$

(a) Find the two values of a for which S does not have a unique solution.

(4)

(b) In the case when $a = 2$, determine the value of b for which S has infinitely many solutions.

(4)
(Total 8 marks)

Q5.

(a) Determine the two values of k for which the system of equations

$$\begin{aligned} x - 2y + kz &= 5 \\ (k+1)x + 3y &= k \\ 2x + y + (k-1)z &= 3 \end{aligned}$$

does not have a unique solution.

(4)

(b) Show that this system of equations is consistent for one of these values of k, but is inconsistent for the other.

(You are not required to find any solutions to this system of equations.)

(8)
(Total 12 marks)

Q6.

The line L and the plane Π have vector equations

$$\mathbf{r} = \begin{bmatrix} 7 \\ 8 \\ 50 \end{bmatrix} + t\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \text{ and } \mathbf{r} = \begin{bmatrix} -2 \\ 0 \\ -25 \end{bmatrix} + \lambda\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} + \mu\begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix}$$

respectively.

(a) Show that L is perpendicular to Π.

(3)

(b) For the system of equations

$$\begin{aligned} 6p + 5q + r &= 9 \\ 2p + 3q + 6r &= 8 \\ -9p + 4q + 2r &= 75 \end{aligned}$$

form a pair of equations in p and q only, and hence find the unique solution of this system of equations.

(5)

(c) It is given that L meets Π at the point P.

(i) Demonstrate how the coordinates of P may be obtained from the system of equations in part (b).

(2)

(ii) Hence determine the coordinates of P.

(2)
(Total 12 marks)

Q7.

Two fixed planes have equations

$$\begin{aligned} x - 2y + z &= -1 \\ -x + y + 3z &= 3 \end{aligned}$$

(a) The point P, whose z-coordinate is λ, lies on the line of intersection of these two planes. Find the x- and y-coordinates of P in terms of λ.

(3)

(b) The point P also lies on the variable plane with equation $5x + ky + 17z = 1$. Show that

$$(k + 13)(2\lambda - 1) = 0$$

(3)

(c) For the system of equations

$$\begin{aligned} x - 2y + z &= -1 \\ -x + y + 3z &= 3 \\ 5x + ky + 17z &= 1 \end{aligned}$$

determine the solution(s), if any, of the system, and their geometrical significance in relation to the three planes, in the cases:

(i) $k = -13$;

(ii) $k \neq -13$.

(6)
(Total 12 marks

Q8.

A system of equations is given by

$$\begin{aligned} x + 3y + 5z &= -2 \\ 3x - 4y + 2z &= 7 \\ ax + 11y + 13z &= b \end{aligned}$$

where a and b are constants.

(a) Find the unique solution of the system in the case when $a = 3$ and $b = 2$.

(5)

(b) (i) Determine the value of a for which the system does not have a unique solution.

(3)

(ii) For this value of a, find the value of b such that the system of equations is consistent.

(4)
(Total 12 marks)

Q9.

Three planes have equations

$$\begin{aligned} x + y - 3z &= b \\ 2x + y + 4z &= 3 \\ 5x + 2y + az &= 4 \end{aligned}$$

where a and b are constants.

(a) Find the coordinates of the single point of intersection of these three planes in the case when $a = 16$ and $b = 6$.

(5)

(b) (i) Find the value of a for which the three planes do not meet at a single point.

(3)

(ii) For this value of a, determine the value of b for which the three planes share a common line of intersection.

(5)

(Total 13 marks)

Q10.

Show that the system of equations

$$x + 2y - z = 0$$

$$3x - y + 4z = 7$$

$$8x + y + 7z = 30$$

is inconsistent.

(Total 4 marks)

Q11.

Consider the following system of equations, where k is a real constant:

$$\begin{aligned} kx &+ 2y &+ z &= 5 \\ x &+ (k+1)y &- 2z &= 3 \\ 2x &- ky &+ 3z &= -11 \end{aligned}$$

(a) Show that the system does not have a unique solution when $k^2 = 16$.

(3)

(b) In the case when $k = 4$, show that the system is inconsistent.

(4)

(c) In the case when $k = -4$:

(i) solve the system of equations;

(5)

(ii) interpret this result geometrically.

(1)

(Total 13 marks)

Q12.

(a) Show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a - b)(b - c)(c - a)$$

(5)

(b) (i) Hence, or otherwise, show that the system of equations

$$x + y + z = p$$

$$3x + 3y + 5z = q$$

$$15x + 15y + 9z = r$$

has no unique solution whatever the values of p, q and r.

(2)

(ii) Verify that this system is consistent when $24p - 3q - r = 0$.

(2)

(iii) Find the solution of the system in the case where $p = 1$, $q = 8$ and $r = 0$.

(5)

(Total 14 marks)

Q13.

A set of three planes is given by the system of equations

$$x + 3y - z = 10$$

$$2x + ky + z = -4$$

$$3x + 5y + (k - 2)z = k + 4$$

where k is a constant.

(a) Show that $\begin{vmatrix} 1 & 3 & -1 \\ 2 & k & 1 \\ 3 & 5 & k-2 \end{vmatrix} = k^2 - 5k + 6$.

(2)

(b) In each of the following cases, determine the **number** of solutions of the given system of equations.

(i) $k = 1$.

(ii) $k = 2$.

(iii) $k = 3$.

(7)

(c) Give a geometrical interpretation of the significance of each of the three results in part (b) in relation to the three planes.

(3)
(Total 12 marks)

Q14.

(a) Show that $(a + b + c)$ is a factor of

$$\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

Express this determinant as the product of $(a + b + c)$ and a quadratic factor.

(5)

(b) Hence, or otherwise, show that there is a single real value of a for which the system of equations

$$ax + 3y + z = -5$$

$$4x + (1 + a)y + (a + 3)z = 9$$

$$2x + (1 - a)y + (a - 3)z = 15$$

does not have a unique solution, and find this value of a.

(5)
(Total 10 marks)